

The Mountain Goats, Every Night

Scraping and modeling 34 years of live setlists from themountaingoats.fandom.com

The dataset

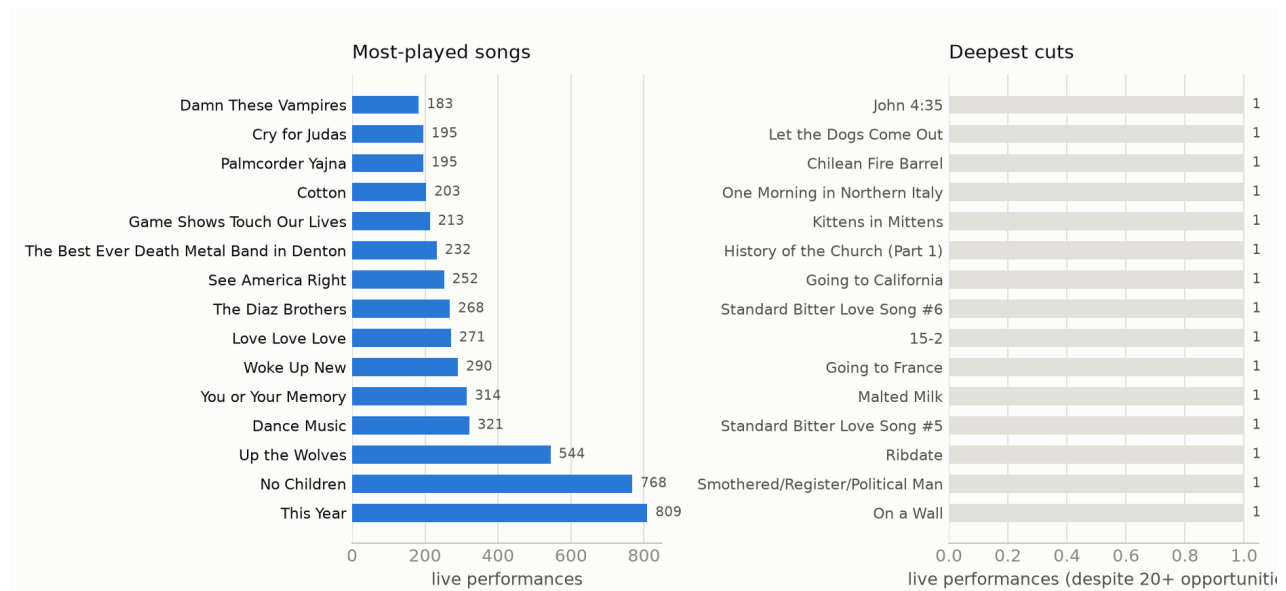
The wiki lists 1,579 live shows from 1992-05-31 through 2026-05-26 — solo John Darnielle sets, full-band tours, radio sessions, festival slots, and Extra Glenns/Extra Lens side-project shows. 1,317 of those have a documented setlist: 22,699 individual song performances across 845 distinct songs, spanning 97 named tours. 1,739 performances have at least one linked YouTube/Vimeo video.

Getting from a wiki category page to those numbers took a few real fixes:

- **Tour isn't in the page text — it's a wiki category** (e.g. *Peter Hughes Farewell Tour 2024*). Two earlier scraper attempts both missed this and scraped a null tour for every show.
- **Encores aren't marked in the setlist table.** They're stated in prose in the Notes section — “The encore was songs 19 through 23” — referencing the table's printed order numbers. A small parser turns ~30 recurring phrasings of that sentence into a per-song encore number.
- **Song identity needed real normalization.** The wiki links each song to its own page, but the same song shows up under underscore-joined slugs, space-joined display text, and case variants of both. Left unmerged, this silently splits a song's play count across rows — a real bug caught mid-project, where “Southwood Plantation Road” initially showed 1 play instead of its real count, 165.
- **Covers needed their own source of truth.** Sniffing setlist notes for the word “cover” only catches about a dozen of them — the wiki actually maintains its own Category:Covers, which tags 156 of the songs played live here. Every song carries an `is_cover` flag from that category, and the prediction model excludes covers from its candidate universe entirely.

The pipeline is two-stage: **fetch** downloads and locally caches every wiki page via the MediaWiki API (resumable, incremental on later runs), and **build** parses the cached HTML into tidy tables entirely offline.

What gets played, and what doesn't

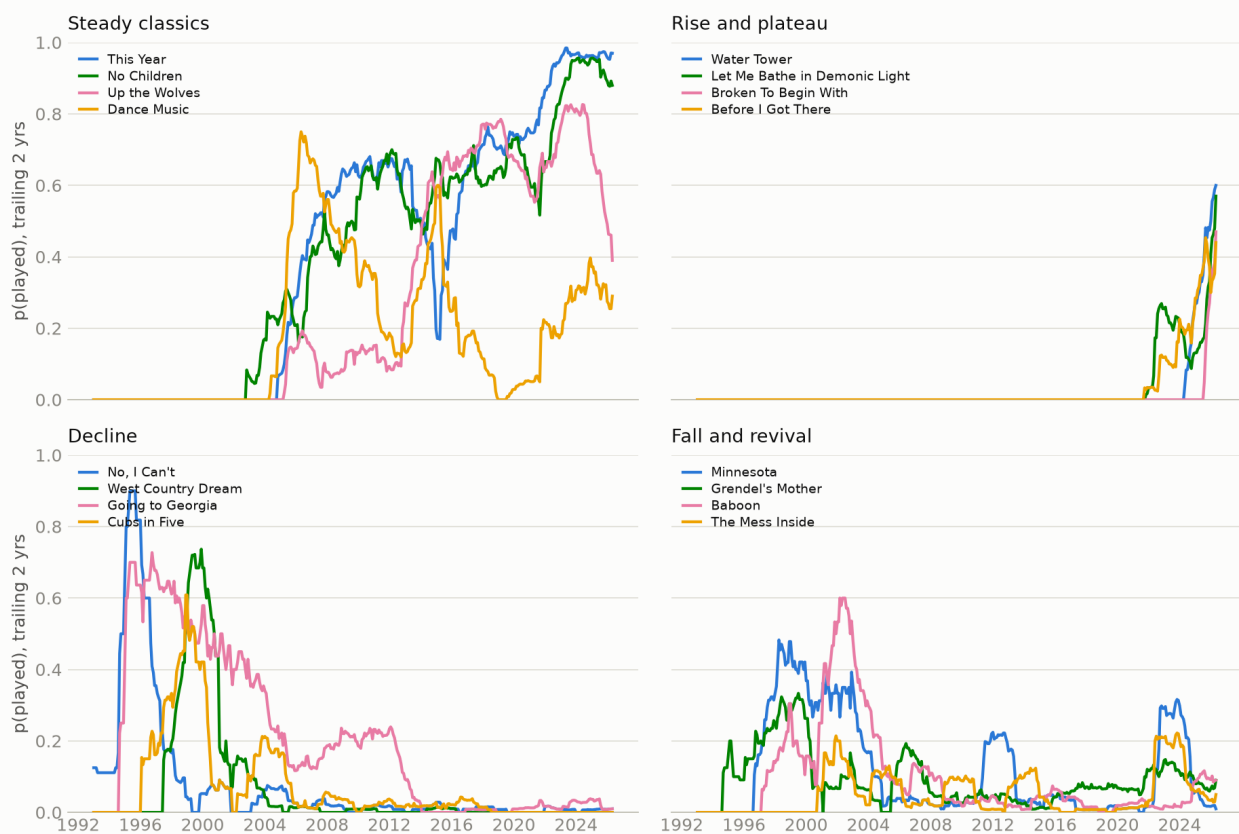


“This Year” and “No Children” anchor the top of the list — both played at a majority of all shows since entering rotation. The deep-cuts panel isn’t just least-played-ever (most of the 845 songs have only 1–2 plays and aren’t interesting on their own); it’s the bottom of an opportunity-adjusted, shrinkage-smoothed play rate among songs with at least 20 chances to be played since their live debut — so a song only counts as a deep cut if it’s been consistently passed over, not just recently written.

Song popularity over time

Treating each song’s live history like a word’s frequency in Google Ngrams — a trailing 2-year play rate sampled monthly — turns 34 years of setlists into trajectories: songs sit at zero before they’re written, rise as they enter rotation, plateau, decline, and sometimes come back.

Song popularity over time



These four groups were picked automatically from the computed trajectories, not hand-selected: the most-played songs overall (steady classics); songs that debuted in the back half of the catalog's history and have climbed straight into rotation since (rise and plateau — a song appearing from nothing and taking hold); the biggest gap between early-career peak and recent play rate (decline); and songs with a real trough between two active periods (fall and revival).

The decline panel caught something the wiki itself corroborates: “Going to Georgia” was a fixture through the mid-2000s, then drops off hard — a 2012 show note has John Darnielle explaining, mid-show, that he considers it a “bullshit song” with an “asshole” narrator and doesn’t want to play it again. The data shows the falloff independent of that anecdote; the anecdote just explains it.

Predicting the setlist

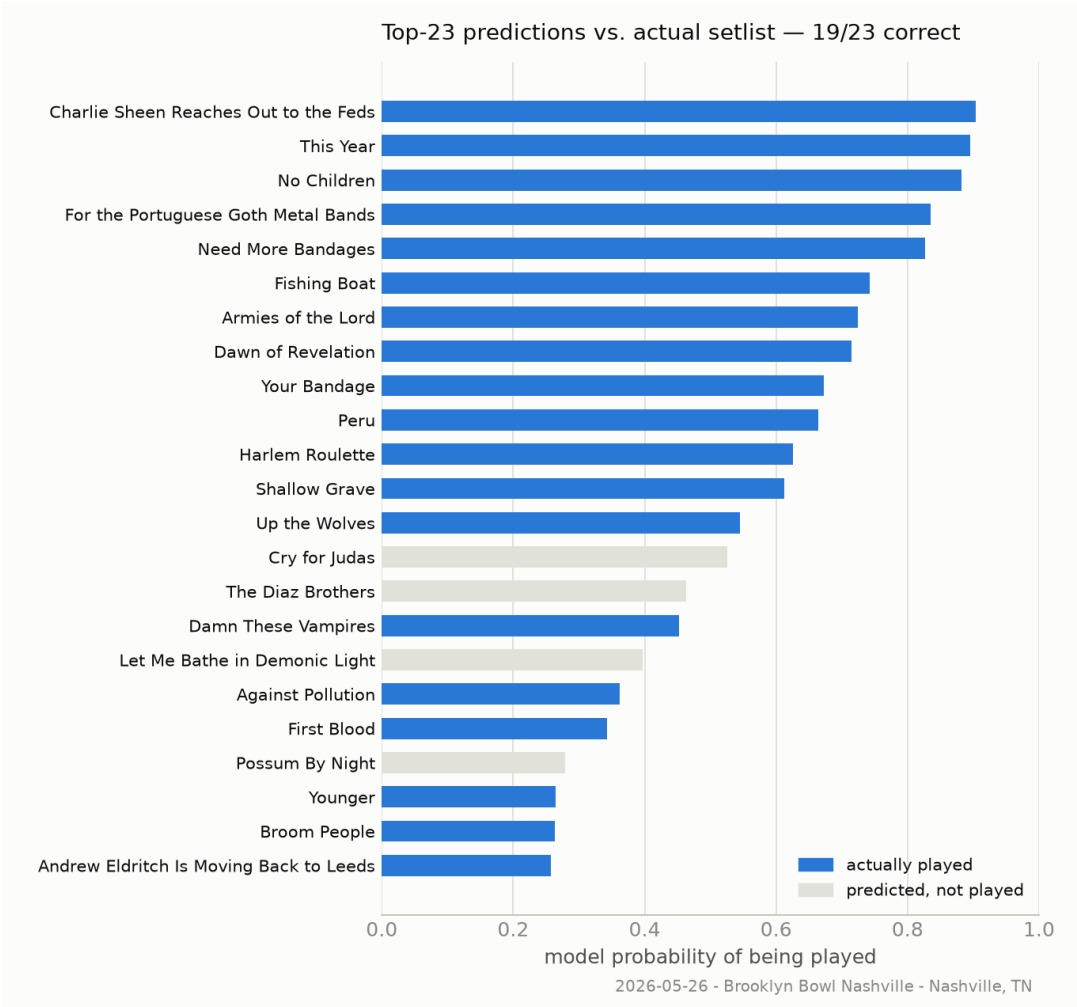
Framed as one binary outcome per (show, song) pair — is this the night song X gets played? — every one of the 1,310 dated, setlisted shows becomes a training example, using only information available before that show: how recently and how often the song's been played, how it's fared on the current tour, its age, and whether it's a shortened-set special appearance (radio/festival/TV).

A logistic regression on those features is compared against a simple but strong baseline — an exponentially-decayed play rate alone — on a strict temporal split (train through 2022, test on the 207 shows from 2023 onward):

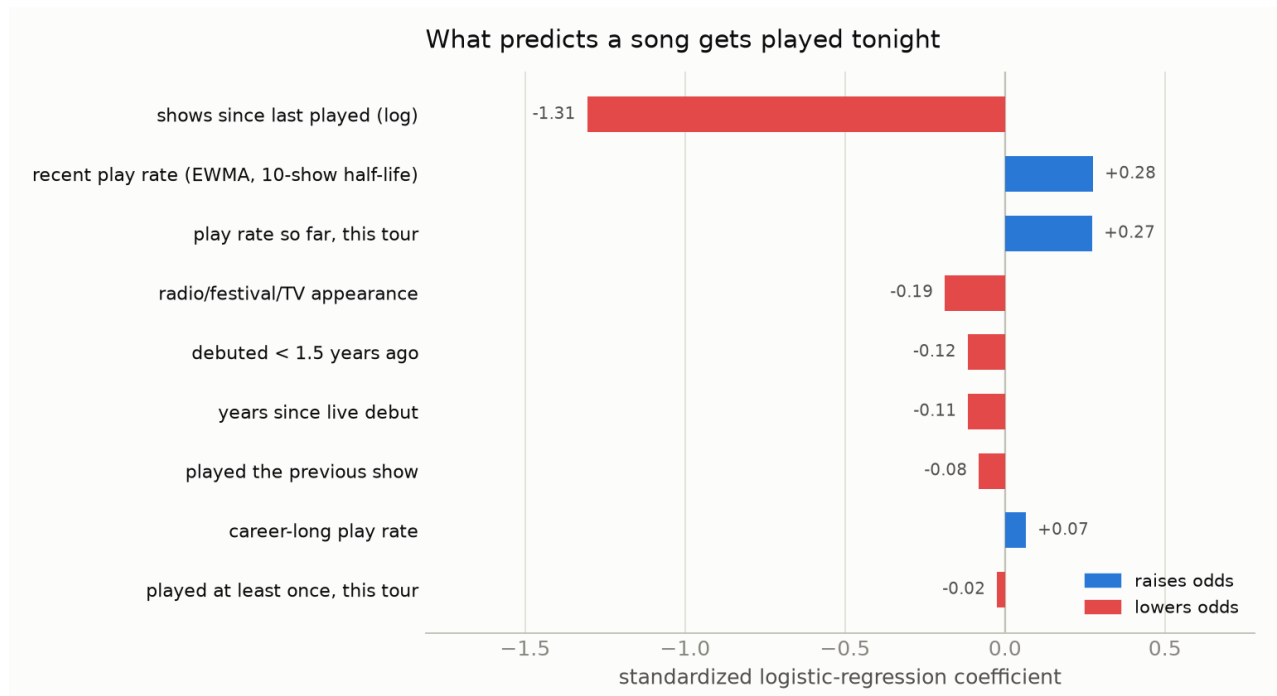
model	log-loss	Brier score	top-n setlist recovery
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Baseline (decayed play rate)	0.0822	0.0195	51.1%
Logistic regression	0.0673	0.0166	59.4%

Top-n setlist recovery: for each show, take the model's n highest-probability songs, where n is the actual number played that night, and measure what fraction were right. On the most recent show in the data:



What actually predicts a setlist

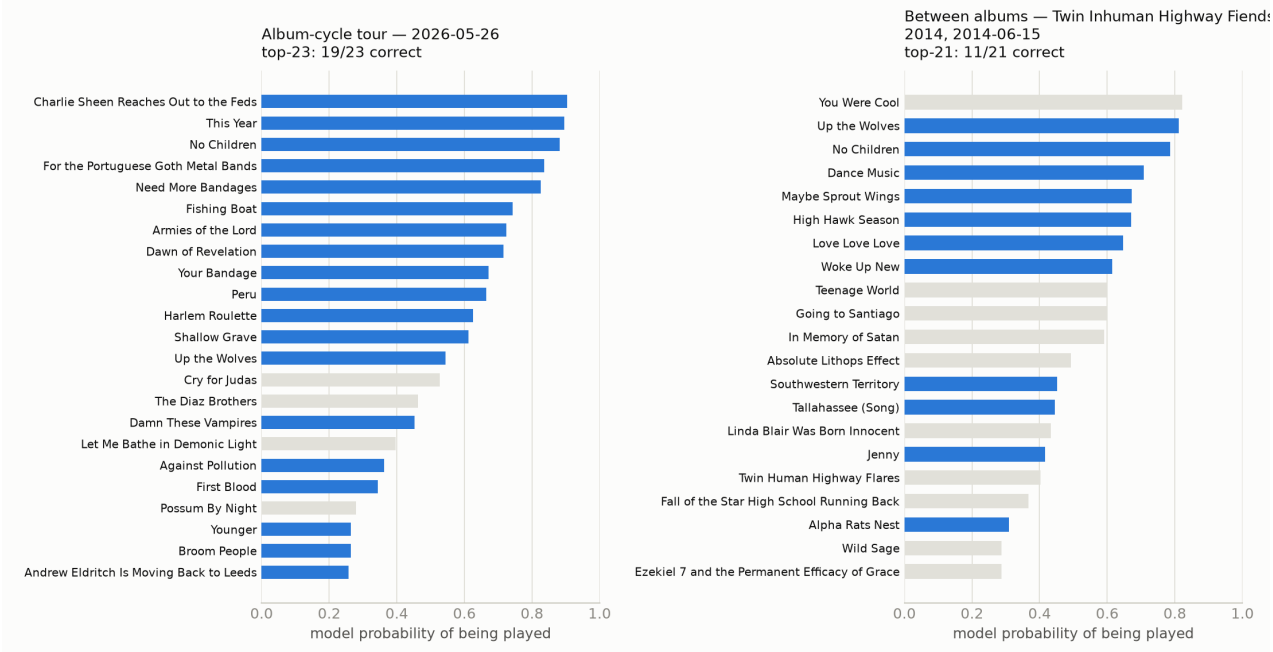


Recency dominates everything else combined — if a song was played a few shows ago, it's overwhelmingly likely to come back soon. That's the visible mechanism behind the tour "rotation": John Darnielle appears to work from a live pool of songs and cycle through it, which is exactly what `tour_rate` (play rate so far, this tour) picking up an independent signal on top of recency confirms. Two smaller, genuinely interesting effects: conditional on recency, brand-new material is actually less sticky than catalog average, and radio/festival/TV appearances reliably favor hits over deep cuts.

Is this just tracking album hype?

The 2026 example above is a new-album tour, and `new_material` is a real feature in the model — worth checking whether the strong recovery number is mostly "predict whatever's newest" in disguise. Among all 2014 tours with at least 8 shows, the one whose actually-played songs had the lowest share of new material — picked automatically, not by hand — is the Twin Inhuman Highway Fiends Tour 2014, squarely between Transcendental Youth (2012) and Beat the Champ (2015):

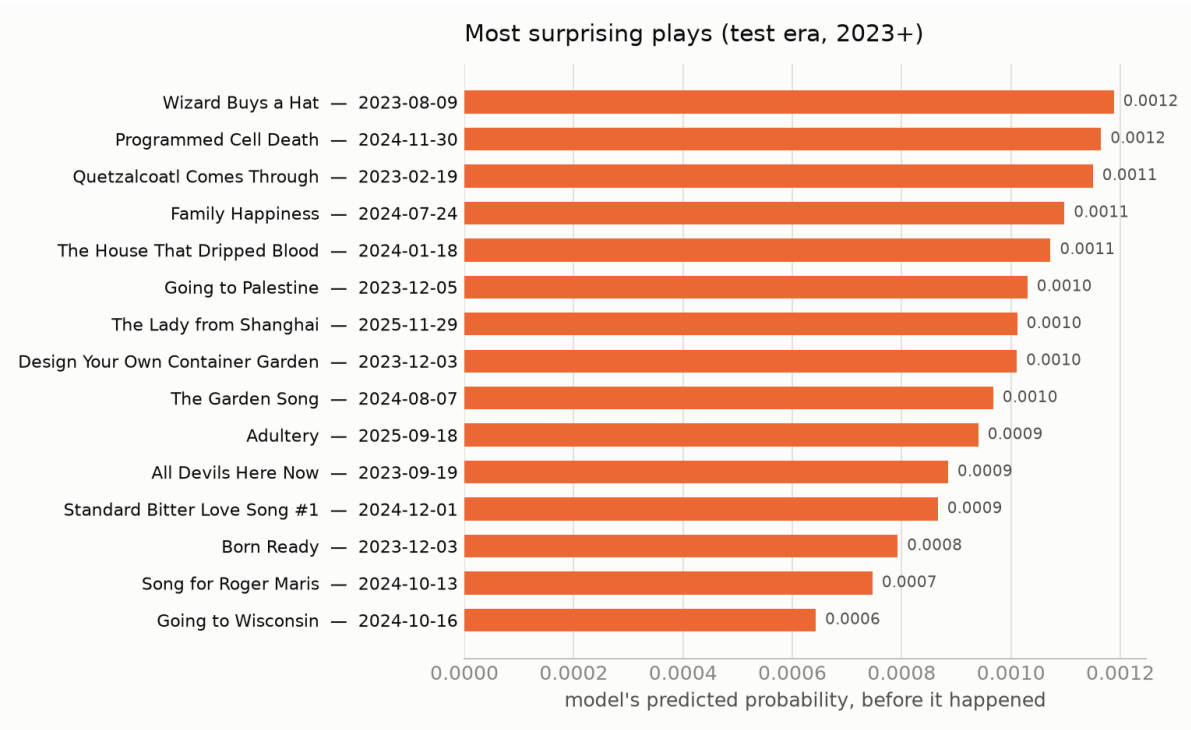
Same model, an album-cycle tour vs. a between-albums tour



The between-albums tour is visibly harder to predict — 55.1% top-n recovery tour-wide, versus 59.4% for the full 2023+ test era — despite 2014 being in-sample (which should make it look artificially easier, not harder). Without a record to promote, the setlist draws more evenly across a wider pool of similarly-loved catalog songs, so there's genuinely more entropy to predict. The model isn't just riding hype; if anything, album cycles are the easy case.

When the model gets surprised

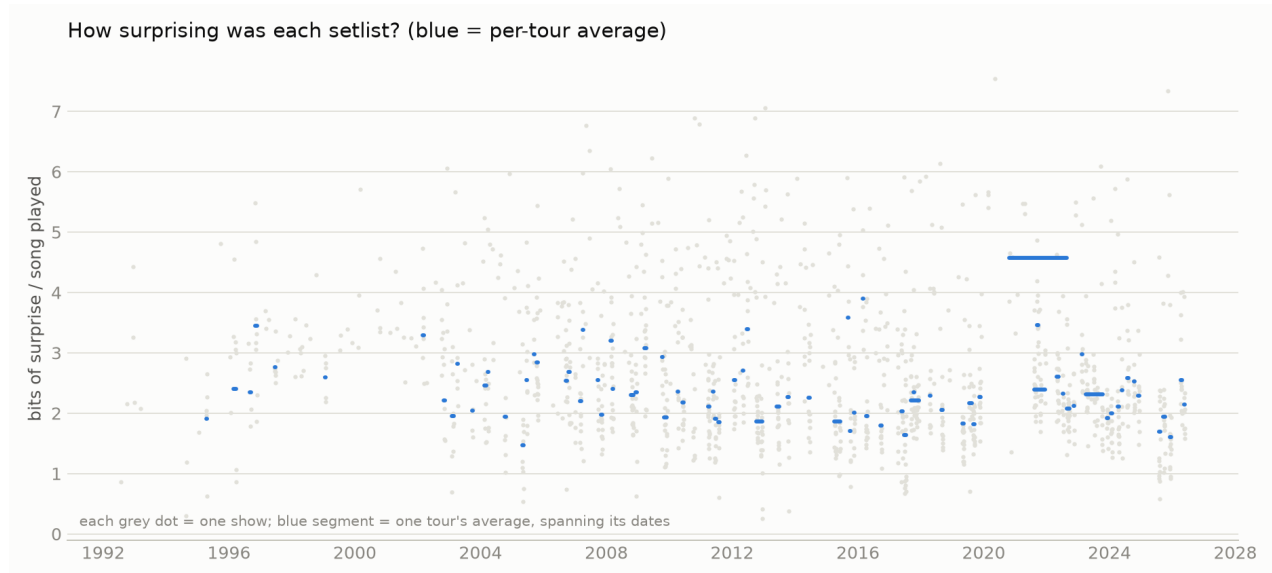
The flip side of a good model is a good list of its misses — nights the model gave a song almost no chance, and it got played anyway:



These are basically a machine-generated “rarities and one-offs” list: songs that came back from years of dormancy for a single night, often at a show with some specific occasion.

How surprising was each setlist as a whole?

Zooming out from individual songs: the same pre-show probabilities give a surprise score for an entire night — the average bits of information across the songs actually played. Aggregated per tour (touring is bursty, so a fixed show-count window saws at tour boundaries):



The tours with the highest average surprise are almost all solo, stripped-down tours — *Winter Solo Tour 2024*, the *Ghost Cave Incubation Chamber solo non-tour*, *All Roads Lead to Lincoln Solo Mini-Tour* — which makes sense: a solo acoustic set draws from a different pool than a full-band show, and the model has no explicit “solo show” feature to account for it. A concrete lead for the next modeling iteration.

Caveats

- The wiki is fan-maintained and lists most, not all, shows — coverage is noticeably better from the mid-2000s on.
- The model has no access to anything outside setlist history — no lyrics, no album themes, no explicit “songs currently being toured.” `tour_rate` and `song_age` are proxies for that.
- Song identity is deduplicated via wiki link slugs where available; a handful of never-linked rarities may still have unmerged spelling variants.
- The model has no “solo show” feature, which the surprise analysis suggests it should — solo/stripped-down tours are consistently hardest to predict.

Reproducing this

```
PY=/Users/bdoughty/opt/miniconda3/envs/tmg-scrape/bin/python
$PY scrape.py fetch && $PY scrape.py build # refresh the raw data
$PY analyze.py # deep cuts, tour summaries
$PY predict.py # the setlist model
$PY timeseries.py # popularity-over-time series
$PY plot_report.py # this report's figures
```

Full data dictionary in README.md; wiki-scraping gotchas and repo layout in CLAUDE.md.